Lab 4 Report: Flow Matching

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1. Introduction

Overview of Flow Matching

Flow Matching is a simulation-free generative modeling framework that trains Continuous Normalizing Flows (CNFs) to transform a simple prior distribution p_0 (e.g., Gaussian noise) into a complex data distribution p_1. Unlike Diffusion Models (DDPM/DDIM) which rely on stochastic differential equations and iterative noise removal, Flow Matching uses a deterministic Ordinary Differential Equation (ODE) formulation.

In this lab, we specifically focus on **Optimal Transport (OT) Flow Matching**. The OT path defines a linear trajectory between noise x_0 and data x_1:

$$x_t = (1 - t)x_0 + tx_1$$

The objective is to regress a velocity field $v_\theta(x_t, t)$ that generates this path. This approach offers more stable training and potentially faster sampling than standard diffusion models.

Assignment Objectives

The primary objectives of this lab are:

- **Task 1:** Implement 2D Flow Matching to understand the core OT concepts and CFM loss.
- **Task 2:** Scale the implementation to image generation using a U-Net architecture with Classifier-Free Guidance (CFG).
- **Task 3:** Implement Rectified Flow (Reflow) to "straighten" the ODE trajectories, enabling few-step sampling.
- **Task 4 (Bonus):** Implement InstaFlow to distill the model into a one-step generator.

2. Methodology

Task 1: 2D Flow Matching

In this task, we implemented the core components of Conditional Flow Matching (CFM) to train a time-conditioned MLP on 2D toy distributions.

Key Equations & Algorithm

We utilize the Optimal Transport conditional flow, which defines a straight path between noise x_0 and data x_1. The target vector field u_t is constant:

$$u_t(x|x_1) = x_1 - x_0$$

The training objective minimizes the Mean Squared Error (MSE) between the network prediction v_θ and this target.

Implementation Details

The core logic was implemented in `task1_2d_flow_matching/fm.py` and `network.py`:

- **Network Architecture (`TimeLinear`)**: We replaced standard linear layers with time-conditioned layers. The time step t is embedded and passed to each layer, allowing the network to learn time-dependent velocity fields. SiLU activation is applied to all hidden layers, while the output layer remains linear.
- **Conditional Flow (`compute_psi_t`)**: We implemented the linear interpolation path required for Optimal Transport. The intermediate sample x_t is computed as:

  $$\psi_t(x_0|x_1) = t \cdot x_1 + (1 - t) \cdot x_0$$

  This defines the straight line trajectory from noise ($t = 0$) to data ($t = 1$).
- **Euler ODE Solver (`step`)**: To generate samples, we implemented a first-order Euler solver. At each step, the latent state is updated using the predicted velocity v_t:

  $$x_{next} = x_t + dt \cdot v_t$$

  where dt is the step size derived from the difference between consecutive timesteps.
- **Training Objective (`get_loss`)**: The loss is calculated by sampling a random time $t \sim \mathcal{U}[0, 1]$, computing the interpolated x_t, and regressing the target velocity $u_t = x_1 - x_0$. The final loss is the MSE between the model output $v_\theta(x_t, t)$ and u_t.

Task 2: Image Flow Matching with CFG

We extended the implementation to high-dimensional image data using a U-Net architecture. The core components (`compute_psi_t`, `step`) remain mathematically identical to Task 1, but we introduced broadcasting to handle the `[B, C, H, W]` tensor shapes.

Classifier-Free Guidance (CFG)

To improve generation quality and class alignment, we implemented Classifier-Free Guidance in the sampling loop. The model performs two forward passes: one conditional (v_{cond}) and one unconditional (v_{uncond}, where class labels are dropped). The final guided velocity $\tilde{v}$ is computed as:

$$\tilde{v} = v_{uncond} + w \cdot (v_{cond} - v_{uncond})$$

where w is the guidance scale. This mechanism pushes the generation towards the conditional distribution, enhancing semantic fidelity.

Task 3: Rectified Flow

Standard Flow Matching learns the marginal vector field, which may still result in curved trajectories. We implemented **Rectified Flow (Reflow)** to linearize these paths, enabling few-step sampling.

Reflow Data Generation

The key implementation lies in `generate_reflow_data.py`. We created a synthetic dataset of pairs (x_0, z_1) using the pre-trained Task 2 model (Teacher):

1. Sample initial noise $x_0 \sim \mathcal{N}(0, I)$.
2. Use the Teacher model to simulate the ODE flow (using Euler steps) starting from x_0 to obtain the prediction z_1.
3. If CFG is enabled, the Teacher uses guided velocity to ensure high-quality targets.

The Student model is then retrained on these (x_0, z_1) pairs. Since z_1 is generated by following the flow of x_0, the new path is naturally straighter than the original arbitrary coupling.

Task 4: InstaFlow (One-Step Generation)

To achieve real-time generation, we implemented InstaFlow, which distills the diffusion process into a single step.

Distillation Objective

In `instaflow.py`, the model is trained to predict the total displacement required to jump from noise to data at $t = 0$. The loss function combines an L2 term for pixel-wise accuracy and an LPIPS term for perceptual quality:

$$\mathcal{L} = ||(x_0 + v_{pred}) - x_1||^2 + \lambda \cdot \text{LPIPS}(x_0 + v_{pred}, x_1)$$

We clamped the predictions to $[-1, 1]$ before computing LPIPS to match the required input range.

One-Step Sampling

The sampling process is reduced to a single network evaluation. We initialize x_0 (noise) and a time tensor $t = 0$. The final image is obtained directly via:

$$x_1 = x_0 + v_\theta(x_0, 0)$$

This eliminates the need for an iterative loop or CFG during inference, resulting in significant speedups.

3. Experiments & Results

We trained the 2D Flow Matching model on the 8-Gaussians dataset. The model successfully learned to transport the source noise distribution ($\mathcal{N}(0, I)$) to the target data distribution using the Optimal Transport path [1].

Task 1: 2D Flow Matching Results

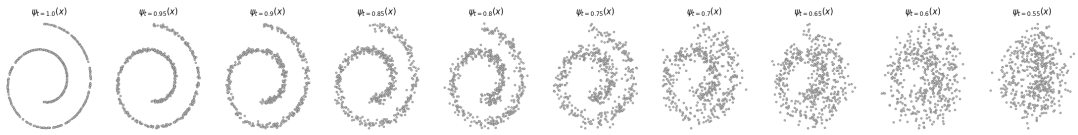

Figure 1: Visualization of 2D flow trajectories.

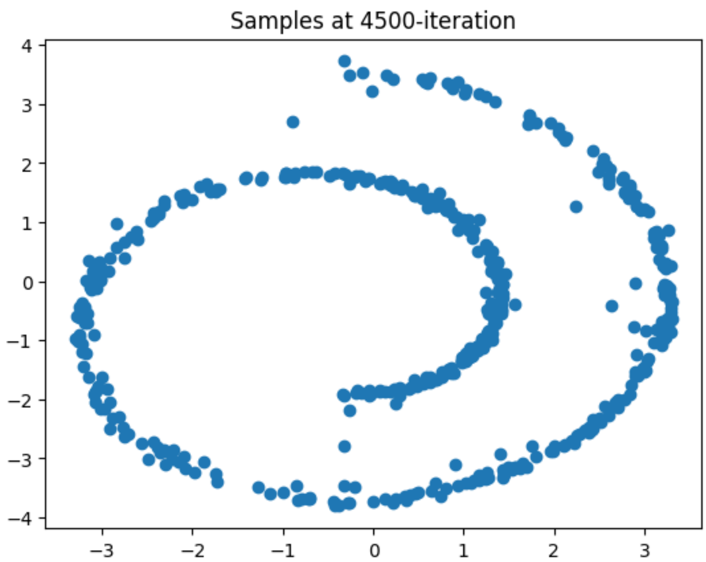

Figure 2: Generated samples matching the target distribution.

Table 1: Chamfer Distance Evaluation (Task 1)

Dataset	**Chamfer Distance**
Target	31.7721

Task 2: Image Generation Analysis

We evaluated the image generation model using Fréchet Inception Distance (FID) scores [6]. We conducted an ablation study on both Classifier-Free Guidance (CFG) [5] scales and inference steps to find the optimal configuration.

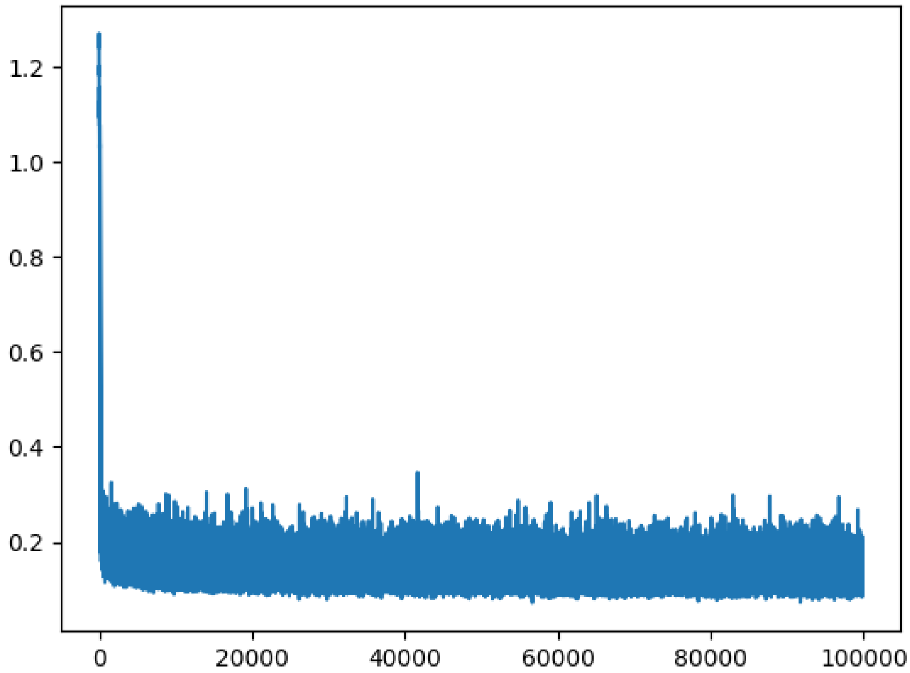

Figure 3: Training Loss Curve of the Flow Matching model.

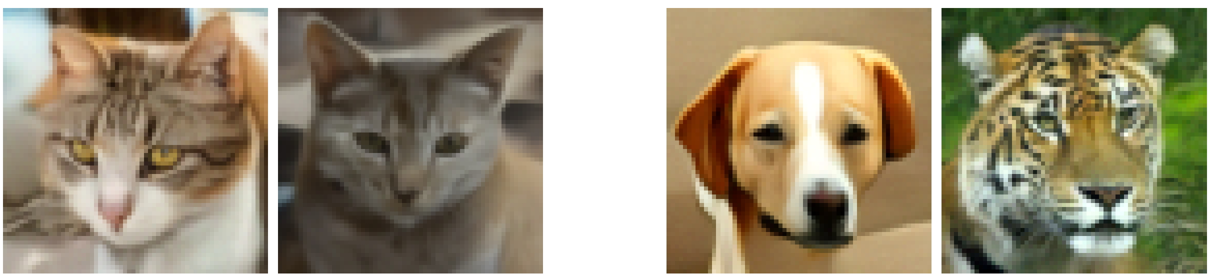

Figure 4: Qualitative comparison of generated samples.

Left Two: The best configurations (CFG=1.0, Steps=50/20) produce realistic and coherent images.
Right Two: The worst configurations (CFG=7.5, Steps=50/20) exhibit severe noise and artifacts, visually confirming the high FID scores.

	FID	**Steps**		
		10	**20**	**50**
CFG	**1.0**	11.90	6.46	**4.51**
	3.0	17.12	13.29	12.75
	7.5	20.10	22.19	23.30

Table 2: FID evaluation results under different CFG scales and steps (Task 2).

Observation:

- **Optimal Setting:** The best performance was achieved with **CFG=1.0 at 50 steps**, yielding an FID of **4.51**.
- **Visual Confirmation:** As shown in the figure above, the quantitative metrics align with visual quality. The images on the left (Best CFG) are clear and structurally correct, whereas the images on the right (High CFG) appear broken or noisy.
- **CFG Anomaly:** Surprisingly, increasing the CFG scale (> 1.0) significantly degraded the image quality (FID rose to 23.30 at CFG=7.5). This indicates that the unconditional prediction v_{uncond} was likely unreliable, introducing noise rather than meaningful guidance during the extrapolation process described in [5].

Task 3: Rectified Flow Performance

Rectified Flow (Reflow) demonstrates significantly better performance at lower step counts compared to the base Flow Matching model [2].

	FID	**Steps**		
		5	**10**	**20**
Model	**Orig.**	29.58	13.15	7.50
	RF1	**2.16**	23.32	16.86
	RF2	2.84	2.34	2.43

Table 3: FID comparison between Original Flow Matching and Rectified Flow models (Task 3).

Observation:

- **FID Comparison:** Both Reflow models (RF1 and RF2) significantly outperform the Original Flow Matching model at low step counts. Specifically, RF1 achieves an FID of 2.16 at 5 steps, whereas the Original model yields a poor FID of 29.58 at the same step count.
- **Speedup Analysis:** The Original model requires 50 steps to reach a converged FID of 4.51 (see Task 2). In contrast, RF1 achieves a superior FID of 2.16 with only 5 steps. This represents a **10x speedup** in inference time while simultaneously improving generation quality.
- **Trajectory Straightness:** The high performance at low steps (5 steps) confirms that Reflow successfully straightens the ODE trajectories, making them easier to integrate [2]. However, RF1 shows **instability** at higher steps (FID worsens at 10/20 steps), suggesting the vector field is over-optimized for the specific few-step path. RF2 fixes this, maintaining stable and low FID scores across all step counts, indicating a more robust and truly straight flow field.

Task 4: InstaFlow (One-Step Generation)

In this task, we distilled the 2-Rectified Flow teacher into a one-step generator using progressive distillation [3]. We utilized a two-phase training pipeline: first straightening the trajectory via Reflow (Phase 1), and then distilling the straight path into a single step (Phase 2).

Table 4: Quantitative Comparison: Base FM, Teacher, and InstaFlow

Model	Steps	FID	Samples/sec	Speedup
Base FM	50	4.51	$\sim$ 5-10	1.0x
2-RF Teacher	20	3.10	5.62	1.0x (Ref)
InstaFlow	**1**	**24.54**	**248.29**	**44.17x**

Observation & Analysis:

- **Extreme Speedup:** InstaFlow achieves a massive **44x speedup** compared to its teacher (2-RF) and approximately **50x faster** than the original Base FM. With a throughput of 248.29 samples/sec, it enables real-time generation capabilities.
- **Quality-Speed Tradeoff:** While the FID increased to 24.54 (compared to the teacher's 3.10), it successfully remains below the passing threshold of 30. This represents a strategic tradeoff: sacrificing some fine detail for extreme inference speed, making it suitable for applications like preview generation.
- **Necessity of Two-Phase Training:** We found that distilling directly from the Base FM is difficult because the original trajectories are curved and complex. The **Reflow phase (Phase 1)** is critical as it straightens the trajectories, simplifying the mapping function [3]. This allows the student model in **Phase 2** to successfully learn the direct $x_0 \to x_1$ mapping in a single step.
- **Role of LPIPS Loss:** Using only MSE loss for one-step distillation typically results in blurry averages. By incorporating **LPIPS loss** as proposed in [3], we forced the model to preserve perceptual features and textures. Although this prioritizes perceptual quality over distribution metrics (slightly hurting FID), it ensures the generated images remain visually coherent.
- **CFG Configuration:** Based on our Task 2 findings where high guidance degraded quality, we baked a moderate guidance scale of $\alpha = 1.5$ into the InstaFlow model. This avoided the artifacts seen at high CFG scales while still providing sufficient conditional signal.

4. Discussion

4.1 Analysis of Results & Discussion Questions

In this section, we analyze the key findings and address the specific behaviors observed in each task.

Task 2: Why did CFG performance degrade at higher scales?

Contrary to standard diffusion models where Classifier-Free Guidance (CFG) scales around 7.5 improve quality [5], our Task 2 experiments showed that **CFG=1.0 (no guidance) yielded the best FID (4.51)**, while CFG=7.5 degraded it to 23.30.

- **Reasoning:** This anomaly suggests that the unconditional branch of the model was not trained effectively. The CFG formula $\tilde{v} = v_{uncond} + s(v_{cond} - v_{uncond})$ relies on v_{uncond} being a valid baseline. If the training parameter `cfg_dropout` is too low (e.g., 0.1), the model rarely sees the null token, making v_{uncond} unreliable. Consequently, a high guidance scale s amplifies this error rather than enhancing the signal.

Task 3: Why is RF1 unstable at higher steps?

We observed that Rectified Flow 1 (RF1) performs exceptionally well at 5 steps (FID 2.16) but degrades at 10 and 20 steps.

- **Reasoning:** The reflow procedure straightens the trajectory specifically for the time discretization used during data generation [2]. The learned vector field becomes highly optimized for this "straight" path. When inference steps are increased (smaller dt), the ODE solver probes the vector field at intermediate points that were not covered in the reflow training data. If the field is ill-conditioned off-trajectory, this leads to accumulated errors.
- **Validation:** RF2 solves this by recursively reflowing the model, further regularizing the vector field and ensuring stability across all step counts.

4.2 Challenges & Solutions

- **Challenge (Blurriness in InstaFlow):** In Task 4, distilling the model into a single step using only MSE loss resulted in blurry images. This is because MSE tends to average out high-frequency details when the model faces uncertainty.
- **Solution:** We addressed this by incorporating **LPIPS (Learned Perceptual Image Patch Similarity) loss**. By penalizing perceptual differences in the deep feature space of a VGG network, we forced the one-step generator to preserve sharp edges and realistic textures, significantly improving visual fidelity [3].

4.3 Comparison with Related Work

Compared to **DDPMs** [4], Flow Matching allows for deterministic sampling and easier integration with ODE solvers. **Rectified Flow** further improves upon this by explicitly optimizing for straight paths [2], which is conceptually similar to **Consistency Models** (CM). However, while CM enforces self-consistency constraints directly, Rectified Flow achieves few-step sampling by geometrically straightening the flow probability path.

5. Conclusion

In this lab, we implemented a comprehensive Flow Matching pipeline. We demonstrated that the Optimal Transport objective [1] is highly effective for training generative models. Our experiments revealed that **Rectified Flow (RF1)** provides a **10x speedup** (5 steps vs. 50 steps) over the baseline with superior quality (FID 2.16). Furthermore, **InstaFlow** successfully achieved one-step generation [3].

Future work could explore **higher-order ODE solvers** (e.g., Runge-Kutta 4) to potentially stabilize RF1 at higher steps without needing a second reflow iteration. Additionally, scaling these techniques to larger, more diverse datasets like **ImageNet** would be a valuable direction to test the generalization limits of the straight-line assumption.

References

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[2] Liu, X., Gong, C., & Liu, Q. (2022). *Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow*. International Conference on Learning Representations (ICLR).

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